

$(x \in A \cup B) \rightarrow x \in U$. From Defn. 3 of Section 2.1, $\boxed{A \cup B \subseteq U}$.

From Prob-9(a), we know that $\boxed{A \cup \bar{A} = U}$.

Q $\therefore U \subseteq A \cup \bar{A}$. By Defn. 3 from Sec-2.1, If x is any arbitrary element of U and $x \in U$, then $x \in A \cup \bar{A}$. From Defn. 1, $x \in A \vee x \in \bar{A}$. If $x \in \bar{A}$, we are done. If $x \in A$, By Defn. 3 from Sec-2.1, $x \in B$. By Defn. of OR, $(x \in \bar{A} \vee x \in B)$. By Defn. 1, $\boxed{x \in \bar{A} \cup B}$.

From Defn. 3 of Sec-2.1, $\boxed{U \subseteq \bar{A} \cup B}$. $\therefore \bar{A} \cup B = U$ $\square$

19) Show that if A, B and C are sets, then $\overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$

a) by showing each side is a subset of the other side

b) using a membership table.

Solution: a) Let, $x \in A \cap B \cap C$. By Defn. 5, $x \notin \overline{A \cap B \cap C}$. By Defn. of not belongs to, $\neg(x \in A \cap B \cap C)$ is True. By Defn. 2, $\neg(x \in A \cap B \cap C)$. By 1st De Morgan's Law, of propositions, $\neg(x \in A \cap B) \vee \neg(x \in C)$. By Defn. 2, $\neg(x \in A \cap B) \vee \neg(x \in C)$. By 1st De Morgan's Law of propositions, $\neg(x \in A) \vee \neg(x \in B) \vee \neg(x \in C)$. By Defn. of not belongs to, $(x \notin A) \vee (x \notin B) \vee (x \notin C)$. By Defn. 5, $(x \in \bar{A}) \vee (x \in \bar{B}) \vee (x \in \bar{C})$. By Twice applying Defn. 1, $x \in \bar{A} \cup \bar{B} \cup \bar{C}$. By Defn. 3 from Section 2.1, $\boxed{A \cap B \cap C \subseteq \bar{A} \cup \bar{B} \cup \bar{C}}$.

Let, $x \in \bar{A} \cup \bar{B} \cup \bar{C}$. By Defn. 1, $x \in \bar{A} \vee x \in \bar{B} \vee x \in \bar{C}$. By Defn. 5, $x \notin A \vee x \notin B \vee x \notin C$. By Defn. of not belongs to, $\neg(x \in A) \vee \neg(x \in B) \vee \neg(x \in C)$. By 1st De Morgan's Law, $\neg(x \in A \cap B \cap C)$. By Defn. 2, $\neg(x \in A \cap B \cap C)$. By 1st De Morgan's Law of propositions, $\neg(x \in A \cap B) \vee \neg(x \in C)$. By Defn. 2, $\neg(x \in A \cap B) \vee \neg(x \in C)$. By Defn. of not belongs to, $x \notin A \cap B \cap C$. By Defn. 5, $x \in \overline{A \cap B \cap C}$. By Defn. 3 from Section 2.1,

$\boxed{\bar{A} \cup \bar{B} \cup \bar{C} \subseteq \overline{A \cap B \cap C}}$ $\square$. $\therefore \overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$ $\square$